DELHI PUBLIC SCHOOL, NAGPUR

Answer Key and Marking Scheme

Term 1 Examination 2026 · Science

Maximum Marks: 80 **Pass Mark: 26**

TO THE EXAMINER

1. Award marks strictly according to the value points listed against each question.
2. A correct answer expressed in the student's own words earns full marks.
3. Where a question carries sub-parts, mark each part independently.
4. The chapter and page reference under each answer is where the question came from. Anything marked **from your class notes** came from the notes rather than the textbook, so the page number is theirs and not the book's.

SECTION A — OBJECTIVE

1. [1 mark] Correct option: (b) $CaCO_3 \rightarrow CaO + CO_2$
Chemical Reactions and Equations

2. [1 mark] Correct option: (d) 10
Acids, Bases and Salts

3. [1 mark] Correct option: (b) Mercury
Metals and Non-metals

4. [1 mark] Correct option: (c) 4
Carbon and its Compounds

5. [1 mark] Correct option: (b) hydrochloric acid
Life Processes

6. [1 mark] Correct option: (b) synapse
Control and Coordination

7. [1 mark] Correct option: (b) blue to green
Chemical Reactions and Equations

8. [1 mark] Correct option: (b) baking soda
Acids, Bases and Salts

9. [1 mark] Correct option: (b) Graphite
Metals and Non-metals

10. [1 mark] Correct option: (c) –COOH
Carbon and its Compounds

11. [1 mark] Correct option: (c) mitochondria
Life Processes

12. [1 mark] Correct option: (b) cerebellum
Control and Coordination

13. [1 mark] Correct option: (a) black and white photography
Chemical Reactions and Equations

14. [1 mark] Correct option: (b) $CaOCl_2$
Acids, Bases and Salts

15. [1 mark] Correct option: (b) sodium and potassium
Metals and Non-metals

16. [1 mark] Correct option: (c) Ethene
Carbon and its Compounds

17. [1 mark] Correct option: (a) Both A and R are true, and R is the correct explanation of A.
Life Processes

18. [1 mark] Correct option: (a) Both A and R are true, and R is the correct explanation of A.

Control and Coordination

19. [1 mark] Correct option: (a) Both A and R are true, and R is the correct explanation of A.

Chemical Reactions and Equations

20. [1 mark] Correct option: (a) Both A and R are true, and R is the correct explanation of A.

Acids, Bases and Salts

SECTION B — VERY SHORT ANSWER

21. [2 marks]

Sodium reacts vigorously with oxygen and water; and can catch fire if left in the open

1. Sodium reacts vigorously with oxygen and water — 1.0
2. and can catch fire if left in the open — 1.0

Metals and Non-metals

22. [2 marks]

Compounds with the same molecular formula but different structures; Butane and isobutane (2-methylpropane)

1. Compounds with the same molecular formula but different structures — 1.0
2. Butane and isobutane (2-methylpropane) — 1.0

Carbon and its Compounds

23. [2 marks]

The body is large and cells are far from the surface; Diffusion is too slow to reach all cells

1. The body is large and cells are far from the surface — 1.0
2. Diffusion is too slow to reach all cells — 1.0

Life Processes

24. [2 marks]

The pathway of a reflex action; from receptor to spinal cord to effector

1. The pathway of a reflex action — 1.0
2. from receptor to spinal cord to effector — 1.0

Control and Coordination

25. [2 marks]

It is covered with a layer of magnesium oxide; which would prevent or slow down burning

1. It is covered with a layer of magnesium oxide — 1.0
2. which would prevent or slow down burning — 1.0

Chemical Reactions and Equations

26. [2 marks]

Dry HCl does not release H^+ ions; H^+ ions form only in the presence of water

1. Dry HCl does not release H^+ ions — 1.0
2. H^+ ions form only in the presence of water — 1.0

Acids, Bases and Salts

SECTION C — SHORT ANSWER

27. [3 marks]

Na loses one electron: $Na \rightarrow Na^+ + e^-$; Cl gains one electron: $Cl + e^- \rightarrow Cl^-$; The oppositely charged ions attract to form Na^+Cl^-

1. Na loses one electron: $Na \rightarrow Na^+ + e^-$ — 1.0
2. Cl gains one electron: $Cl + e^- \rightarrow Cl^-$ — 1.0
3. The oppositely charged ions attract to form Na^+Cl^- — 1.0

Metals and Non-metals

28. [3 marks]

Reaction of an alcohol with a carboxylic acid to form an ester; $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (acid catalyst); Esters are used in perfumes and flavouring agents

1. Reaction of an alcohol with a carboxylic acid to form an ester — 1.0
2. $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (acid catalyst) — 1.0
3. Esters are used in perfumes and flavouring agents — 1.0

Carbon and its Compounds

29. [3 marks]

Digested food passes into blood through the intestinal wall; Villi increase the surface area for absorption; Villi have a rich blood supply that carries absorbed food away

1. Digested food passes into blood through the intestinal wall — 1.0
2. Villi increase the surface area for absorption — 1.0
3. Villi have a rich blood supply that carries absorbed food away — 1.0

Life Processes

30. [3 marks]

Dendrites receive information; The cell body processes it; The axon carries the impulse to the next neuron

1. Dendrites receive information — 1.0
2. The cell body processes it — 1.0
3. The axon carries the impulse to the next neuron — 1.0

Control and Coordination

31. [3 marks]

A more reactive element displaces a less reactive one from its compound; $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$; The blue solution turns green and a brown coating forms on iron

1. A more reactive element displaces a less reactive one from its compound — 1.0
2. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ — 1.0
3. The blue solution turns green and a brown coating forms on iron — 1.0

Chemical Reactions and Equations

32. [3 marks]

Reaction of an acid with a base to give salt and water; $NaOH + HCl \rightarrow NaCl + H_2O$; Antacids neutralise excess stomach acid

1. Reaction of an acid with a base to give salt and water — 1.0
2. $NaOH + HCl \rightarrow NaCl + H_2O$ — 1.0
3. Antacids neutralise excess stomach acid — 1.0

Acids, Bases and Salts

33. [3 marks]

Roasting: heating a sulphide ore strongly in excess air; Calcination: heating a carbonate ore strongly in limited air; $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnCO_3 \rightarrow ZnO + CO_2$

1. Roasting: heating a sulphide ore strongly in excess air — 1.0
2. Calcination: heating a carbonate ore strongly in limited air — 1.0
3. $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnCO_3 \rightarrow ZnO + CO_2$ — 1.0

Metals and Non-metals

SECTION D — LONG ANSWER

34. [5 marks]

Ethanol is a liquid that is soluble in water and burns with a clean flame; Ethanoic acid is a weak acid; it turns blue litmus red and freezes at 290 K; Sodium hydrogencarbonate gives brisk effervescence (CO_2) with ethanoic acid; Ethanol gives no effervescence with it; Drinking alcohol slows reactions and damages the liver

1. Ethanol is a liquid that is soluble in water and burns with a clean flame — 1.0
2. Ethanoic acid is a weak acid; it turns blue litmus red and freezes at 290 K — 1.0
3. Sodium hydrogencarbonate gives brisk effervescence (CO_2) with ethanoic acid — 1.0
4. Ethanol gives no effervescence with it — 1.0
5. Drinking alcohol slows reactions and damages the liver — 1.0

Carbon and its Compounds

35. [5 marks]

A nephron has a Bowman's capsule containing the glomerulus; Blood is filtered in the glomerulus into the capsule; Useful substances such as glucose, amino acids, salts and water are reabsorbed in the tubule; The remaining fluid becomes urine and passes to the collecting duct; The amount of urine depends on how much excess water and dissolved waste there is

1. A nephron has a Bowman's capsule containing the glomerulus — 1.0
2. Blood is filtered in the glomerulus into the capsule — 1.0
3. Useful substances such as glucose, amino acids, salts and water are reabsorbed in the tubule — 1.0
4. The remaining fluid becomes urine and passes to the collecting duct — 1.0
5. The amount of urine depends on how much excess water and dissolved waste there is — 1.0

Life Processes

36. [5 marks]

Forebrain (cerebrum): thinking, memory and voluntary actions; Midbrain: involuntary actions such as reflex movements of the eye; Hindbrain: cerebellum for balance; medulla and pons for involuntary actions; The skull (cranium) protects it; Cerebrospinal fluid absorbs shocks

1. Forebrain (cerebrum): thinking, memory and voluntary actions — 1.0
2. Midbrain: involuntary actions such as reflex movements of the eye — 1.0
3. Hindbrain: cerebellum for balance; medulla and pons for involuntary actions — 1.0
4. The skull (cranium) protects it — 1.0
5. Cerebrospinal fluid absorbs shocks — 1.0

Control and Coordination

SECTION E — CASE-BASED QUESTIONS

37. [4 marks]

Nitrogen dioxide (NO_2); $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; Thermal decomposition reaction; Lead oxide (PbO)

1. Nitrogen dioxide (NO_2) — 1.0
2. $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ — 1.0
3. Thermal decomposition reaction — 1.0
4. Lead oxide (PbO) — 1.0

Chemical Reactions and Equations

38. [4 marks]

D; B; C; C < D < A < B

1. D — 1.0
2. B — 1.0
3. C — 1.0
4. C < D < A < B — 1.0

Acids, Bases and Salts

39. [4 marks]

Aluminium; Copper; Fe + $CuSO_4$ → $FeSO_4$ + Cu; Al > Zn > Fe > Cu

1. Aluminium — 1.0
2. Copper — 1.0
3. Fe + $CuSO_4$ → $FeSO_4$ + Cu — 1.0
4. Al > Zn > Fe > Cu — 1.0

Metals and Non-metals

END OF ANSWER KEY